

Erdős Problem #367

EP-367

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $B_2(n)$ be the 2-full part of n (that is, $B_2(n) = n/n'$ where n' is the product of all primes that divide n exactly once). Is it true that, for every fixed $k \geq 1$, $\prod_{n \leq m < n+k} B_2(m) \ll n^{2+o(1)}$? Or perhaps even $\ll_k n^2$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>